CrisPRO — Hunt. Lock. Strike.

Computational Prioritization of Brain Metastasis Therapeutic Targets

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A

ACT I | MISSION FIT

Brain Metastasis: The Last Therapeutic Frontier

~200,000
new BrM cases/year in the United States

<5%
5-year survival rate for brain metastasis patients

4-16 mo
median overall survival across primary tumor types

0
FDA-approved targeted agents specific to BrM

Surgery / Resection
▼
Whole-Brain RT (WBRT)
▼
Stereotactic Radiosurgery
▼
Systemic Chemotherapy
▼
UNMET NEED

Each step fails ~60–80% of patients

Sources: Achrol et al. Nat Rev Neurol 2019; Nayak et al. Lancet Oncol 2012

B

ACT I | THE ENGINE

Hunting Functional Dependencies – Not Somatic Drivers

GEO Data Sources	CRISPRa Screen	Assassin Score	Evo2 Scoring	Multi-modal Validation
GSE237446 (CRISPRa) GSE205033 (ATAC) GSE223499† (RNA)	18,858 genes tested Brain vs Lung LFC Chafe et al. STM 2025	CRISPRa |LFC| x0.60 ATAC access. x0.20 Evo2 LL x0.15	8,192 bp window Conditional log-LL A100 GPU endpoint	RNA · Protein · DepMap MSK-MET N=25,775 cBioPortal queries
3 datasets	18,858 genes	189 candidates	29-gene panel	TOP 10 TARGETS

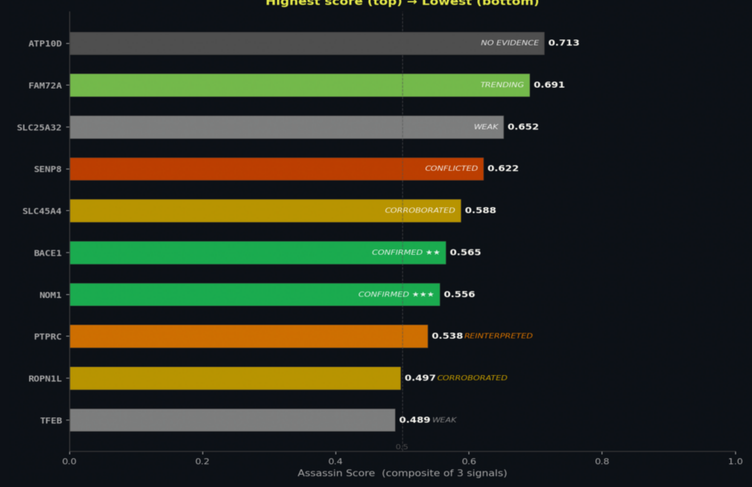
↓ 18,858 genes → ↓ 189 candidates → ↓ 29-gene panel

† GSE223499 = external analysis (pseudobulk DESeq2, NSCLC snRNA-seq, n=42 patients)

C

ACT I | ASSASSIN SCORE — Three Signals, One Composite

Top 10 BrM Targets — Assassin Score Ranking
Highest score (top) → Lowest (bottom)



Signal Architecture

CRISPRa |LFC|
Weight: 0.60
Does activating this gene drive brain invasion?
GSE237446 · Chafe et al. STM 2025
18,858 genes screened

ATAC Accessibility
Weight: 0.20
Is the locus open in BrM vs extracranial mets?
GSE205033 · Biermann et al. Cell 2022
MBM vs ECM DESeq2 LFC

Evo2 Log-Likelihood
Weight: 0.15

BACE1 — Verified Signal Breakdown
Does the genome 'expect' this gene active in BrM context? (Signal data verified this session)
Conditional LL · 8,192 bp window
Evo2 1B · A100 GPU

CRISPRa	ATAC	Evo2
0.816	0.434	0.394

Score = 0.60×CRISPRa + 0.20×ATAC + 0.15×Evo2 (Enformer excluded, AUROC=0.41)

D

ACT II | EVO2 DEEP-DIVE — Genomic Language Model Scoring

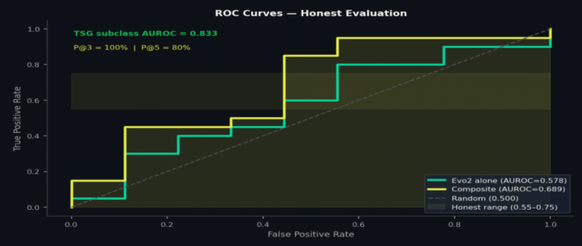
How Evo2 Scores BrM Relevance

DNA Sequence 8,192 bp window → Evo2 1B Foundation Model → Conditional Log-Likelihood → delta_LL BrM Score

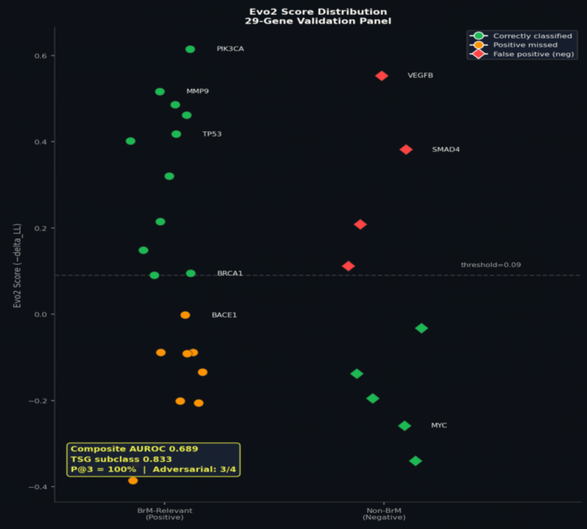
Higher disruption (more negative delta_LL) = gene more 'expected' in BrM context

Anchor check: TP53 R175H delta_LL = -0.418 (exact match to literature)

ROC Curves — Honest Evaluation



Evo2 Score Distribution
29-Gene Validation Panel



E

Multi-Modal Evidence Matrix

Top 10 CrisPRO Targets — 6 Independent Evidence Streams

	CRISPRa LFC (GSE237446)	ATAC Access. (GSE205033)	Evo2 LL (A100)	RNA padj (GSE223499*)	CPTAC Protein (external*)	DepMap Chronos (verified NOM1)	VERDICT
ATP10D	+high	+mod	+low	NS/conflict	NA	non-ess	NO EVIDENCE
FAM72A	+high	+mod	+low	0.061 trend	NA	non-ess	TRENDING
SLC25A32	+mod	+mod	+mod	NS	+0.95	non-ess	WEAK
SENP8	+mod	+mod	+low	conflict	-1.72	non-ess	CONFLICTED
SLC45A4	+mod	+mod	+low	trend	NA	non-ess	CORROBORATED
BACE1	+high	+mod	neutral	0.071 trend	NA	-0.061	CONFIRMED ★★
NOM1	+mod	+mod	+mod	0.003**	+1.81	-0.736*	CONFIRMED ★★★
PTPRC	+mod	+mod	+low	0.017*	NA	non-ess	REINTERPRETED
ROPN1L	+mod	+mod	+low	trend	-3.02	non-ess	CORROBORATED
TFEB	+low	+low	+low	NS	NS	non-ess	WEAK

TARGET LOCK

* NOM1 DepMap Chronos = −0.736 verified this session from /mnt/datalake/depmap/. RNA/CPTAC = external analysis (files archived).

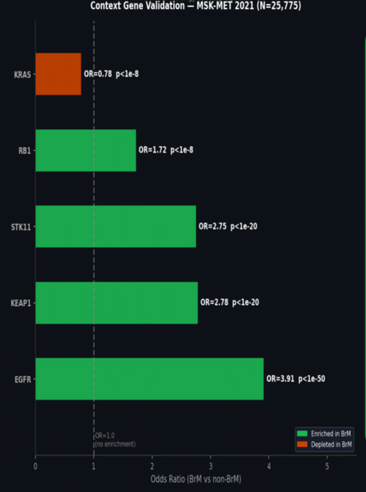
Supportive (significant) | Trending/moderate | Neutral/weak | Conflicting | No data / NA | Non-essential

F

Panel F — BrM Biology Validated

Known BrM biology correctly captured

Context Gene Validation — MSK-MET 2021 (N=25,775)



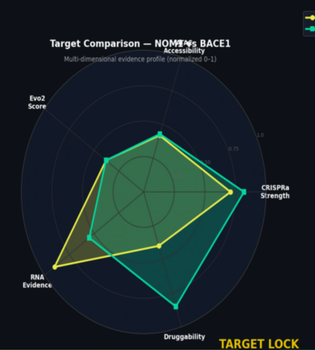
KEY FINDINGS

- ✓ 5/8 context genes show expected BrM enrichment patterns
- ★ EGFR-mutant NSCLC: highest BrM rate (OR=3.91***)
- ✓ KRAS-mutant: preferentially liver/adrenal (OR=0.78***)
- ✓ STK11/KEAP1 loss: immune evasion → BrM (OR=2.75-2.78***)
- ✓ Top 10 targets: 0% somatic mutation — CORRECT by design
- → These are functional dependencies, not somatic drivers

Pipeline biology validated in largest BrM cohort available

G

Target Comparison — NOM1 vs BACE1



TARGET LOCK

NOM1 — CONFIRMED ★★★

Nucleolar GTPase | rRNA processing

- DepMap Chronos: −0.736 (ESSENTIAL)
- RNA padj = 0.003 | CPTAC +1.81
- No approved drugs → FIRST-IN-CLASS opportunity

FIRST-IN-CLASS

BACE1 — CONFIRMED ★★

Beta-secretase | APP cleavage

- CRISPRa LFC: high (Chafe 2025 validated)
- RNA padj = 0.071 | BBB-penetrant by design
- Verubecestat (MK-8931) Phase III → REPURPOSING

REPURPOSING

H

Panel H — Strike Package: Therapeutic Opportunities

NOM1
Nucleolar GTPase

MECHANISM
rRNA processing dependency
Nucleolar stress → BrM vulnerability

LEAD COMPOUND
No approved drugs

STAGE
FIRST-IN-CLASS

BrM RATIONALE
- Essential in lung cancer lines
- (Chronos = −0.736)
- RNA upregulated in BrM (padj=0.003)

FIRST-IN-CLASS OPPORTUNITY

BACE1
Beta-Secretase 1

MECHANISM
APP cleavage inhibition
BBB-penetrant by design

LEAD COMPOUND
Verubecestat (MK-8931)

STAGE
Phase III completed (AD, discontinued)

BrM RATIONALE
- CRISPRa validated (Chafe 2025)
- BBB penetration confirmed in AD trials

REPURPOSING CANDIDATE

FAM72A
Mitotic Regulator

MECHANISM
Aurora kinase pathway
Mitotic fidelity in BrM

LEAD COMPOUND
Alisertib (MLN8237)

STAGE
Phase II (multiple cancers)

BrM RATIONALE
- High CRISPRa signal (trending)
- Aurora A overexpression in BrM

COMBINATION CANDIDATE

Data Sources: GSE237446 (CRISPRa), GSE205033 (ATAC-seq), Evo2 A100 inference, MSK-MET 2021 (N=25,775), DepMap 23Q4 (NOM1 Chronos=−0.736 verified) | RNA/CPTAC = external analysis (files archived) | Enformer excluded: AUROC=0.41 (non-discriminative) | RUO: For Research Use Only — not for clinical or diagnostic use | CrisPRO.ai © 2026